

Assignment #1 Self-Evaluation

Roll-number: _____

Section: _____

Category	Criteria	Marks	Obtained
Input Parsing & Environment Setup		10	
	Correctly parses and maps the grid	5	
	Correctly parses robot parameters (ID, priority, energy)	2	
	Correctly parses start, goal, and checkpoint coordinates	3	
State Space Design		10	
	State representation includes robot position (x, y)	3	
	State tracks checkpoint progress (checkpoint index)	3	
	State tracks time or step count for collision checking	4	
Search Algorithm Implementation		20	
	Correct implementation of Search Algorithm	10	
	Correct generation of successor states (4 moves + wait)	5	
	Proper visited-state management	5	
Movement Constraints & Environment Rules		15	
	Prevents illegal movement (outside grid/into obstacles)	7	
	Correct implementation of one-way cell rules (^, v, <, >)	6	
	Allows waiting in place as a valid action	2	
Checkpoint Handling		10	
	Ensures checkpoints are visited in the required order	6	
	Correctly updates checkpoint index when reached	2	
	Ensures goal state only valid after all checkpoints	2	
Energy & Time Constraints		10	
	Correct time and energy updates per action	6	
	Path length respects robot energy limit	4	
Multi-Robot Priority Planning		15	
	Robots sorted correctly by descending priority	5	
	Paths planned sequentially for each robot	5	
	Reservation table implemented for previous robots	5	
Collision Avoidance		5	
	Prevents two robots occupying same cell at same time	5	
Output File Generation		5	
	Correct output format in output.txt	3	
	Outputs correct error when no valid path exists	2	
TOTAL		100	